

Karthik Vetrivel

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EDUCATION

Stanford University

Sept 2021 - June 2025

B.S./M.S. Computer Science

3.9/4.05 GPA

- Coursework (for B.S. + M.S.): Computer Vision & Deep Learning, Machine Learning, Natural Language Processing, Parallel Computing, Distributed Systems, Operating Systems, Data Structures & Algorithms

EXPERIENCE

NVIDIA

September 2025 – Present

Software Engineer, AI Infrastructure

New York, NY

- Develop GPU Operator (Go, Kubernetes, Helm) — NVIDIA's open-source platform (10k+ GitHub stars) powering GPU provisioning for large-scale LLM training and inference across AWS, GCP, and Azure.
- Designed and shipped ISV self-certification framework, replacing manual validation with an automated suite certifying GPU Operator across production hardware fleets, accelerating enterprise onboarding. Featured at GTC '26.
- Architected zero-downtime GPU driver upgrades via idempotent state management, eliminating 3–5 min of lost GPU compute per node restart across DGX Cloud and 1000+ GPU training clusters.
- Owned cross-stack features spanning driver lifecycle, device scheduling, container runtime, GPU telemetry (DCGM), and the Kubernetes DRA driver for multi-GPU allocation in LLM training clusters.
- **Open Source:** Authored graph fusion transforms and operator pattern matchers in TensorRT-LLM (Python, PyTorch) accelerating inference for Mamba/Bamba SSM and MoE architectures, plus torch.export fixes for FP8 quantized models.

NVIDIA

June 2024 – September 2024

Software Engineer Intern

Santa Clara, CA

- Engineered a custom Kubernetes operator in Go for NVIDIA Cloud Functions (NVCF), orchestrating serverless GPU workloads with automated network provisioning across 100+ compute instances.
- Built multi-threaded state reconciliation API for NVCF control plane in Go, force-syncing desired state across distributed nodes and reducing mean time to resolution by 14%.

Tesla (Autopilot)

January 2024 – March 2024

Machine Learning Engineering Intern

Palo Alto, CA

- Implemented out-of-core distributed GPU training pipeline for XGBoost in Python, scaling to 500GB+ datasets across multi-node GPU clusters with optimized data parallelism and mixed-precision computation.
- Designed and deployed a Mixture-of-Experts (MoE) neural network for automatic park mode activation on Tesla Model 3 & Y, achieving sub-200ms inference latency on embedded vehicle hardware.

Stanford AI Lab

June 2022 – December 2022

Research Intern

Stanford, CA

- Built [dataset](#) and training pipeline for deep learning-based human motion generation. Published in CVPR '23 (top-cited AI/ML venue). Joint Stanford + Meta AI.
- Eliminated networking bottleneck in live neural avatar generation using C and Linux socket-level optimizations, cutting latency from 200ms to 15ms (13x speedup).
- Architected secure AWS S3 pipeline for open-source dataset and model release, featured [here](#).

PROJECTS

Stanford AI Lab & Google Research – Context-Aware 3D Scene Generation

May 2023

- Leveraged LLM prompting and chain-of-thought reasoning to embed commonsense knowledge for robotic 3D scene arrangement, enabling semantically-valid real-world object placement.

TECHNICAL SKILLS

Languages: Python, C, C++, Go, CUDA, Java, Javascript, Typescript, HTML/CSS

ML/AI: PyTorch, TensorFlow, Transformers, Mixed-Precision Training, Distributed GPU Training, Model Quantization (FP8), NCCL, DeepSpeed

Systems: Linux, Kubernetes, Docker, AWS, Azure, GCP, eBPF, Container Networking

Tools: Git, Weights & Biases, Jenkins, CI/CD, Terraform